

Seasonal Trading Strategies: Identifying Optimal Entry and Exit Windows across NEPSE Sub-Sectors

Tek Raj Pant Himal Jung Oli
M.Sc. Computational Mathematics
Computational Statistics

Department of Mathematics
School of Science
Kathmandu University

June 2026

Outline

- 1 Introduction
- 2 Objectives
- 3 Data Source
- 4 Methodology
- 5 Results and Discussion
- 6 Conclusion

Introduction

- The Nepal Stock Exchange (NEPSE) often exhibits patterns influenced by the Bikram Sambat calendar, fiscal year cycles, and major festivals like Dashain and Tihar [1].
- This project explores the *Seasonality Hypothesis* — stock returns are not purely random but follow periodic trends that challenge weak-form efficiency [2].
- By analysing sectoral performance over five years (2021–2026), we distinguish genuine anomalies from noise [3].
- Different sub-sectors show unique trends tied to fiscal year closings and festive windows [4]. Our goal: identify optimal entry and exit windows using historical data [5, 6].

Objectives

- ➊ Identify statistically significant monthly/seasonal trends within NEPSE sub-sectors.
- ➋ Quantify risk-return profiles during high-volatility periods.
- ➌ Develop a Z-score based entry–exit framework for optimising trading timing.

Data Source

- **Period:** 2021 – 2026 (daily closing prices).
- **Source:** Nepse Alpha (<https://nepsealpha.com/nepse-data>).
- **Sectors covered:** Banking, Hydropower, Manufacturing, Microfinance, Non-Life Insurance, Others.
- Daily returns are computed as:

$$R_t = \frac{P_t^{\text{close}} - P_{t-1}^{\text{close}}}{P_{t-1}^{\text{close}}}$$

where P_t^{close} denotes the closing price at time t .

Methodology – Exploratory & Distributional Analysis

- **Boxplots** [7]: IQR-based outlier detection across sectors.
- **Histograms & Q–Q plots** [8]: assess normality departures.
- **Skewness** [9]: $g_1 = \frac{1}{n} \sum \left(\frac{x_i - \bar{x}}{s} \right)^3$.
- **Excess kurtosis** [10]: $g_2 = \frac{1}{n} \sum \left(\frac{x_i - \bar{x}}{s} \right)^4 - 3$.
- **Shapiro–Wilk test** [11]: formal normality test.
- **Additional EDA**: monthly bar charts with bootstrap CIs, density plots with normal reference, cumulative time-series, heatmap of sector×month returns, ACF plots.

Methodology – Confirmatory & Seasonality Detection

Hypothesis Tests

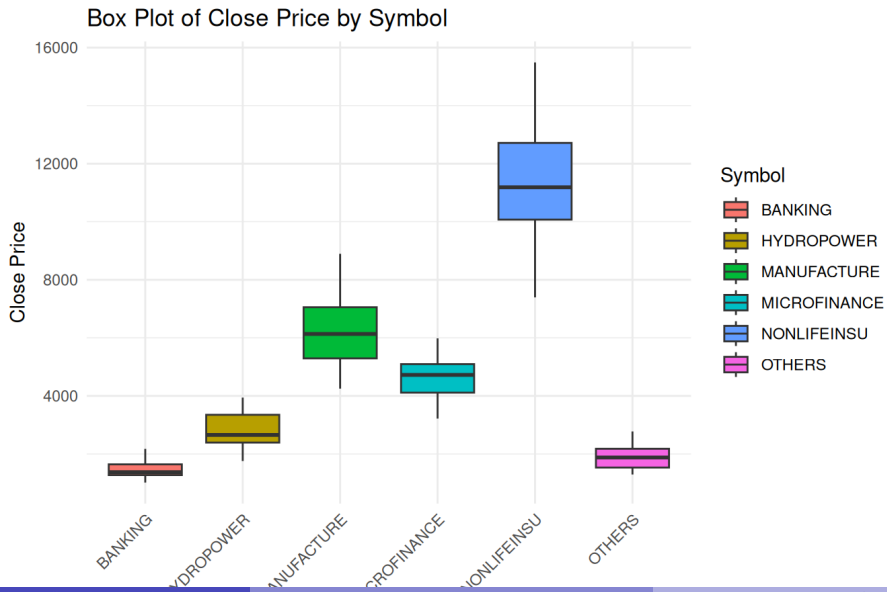
- Chi-square independence [12]
- One-way ANOVA (month & sector) [13]
- Two-way ANOVA (Sector×Month) [14]
- Kruskal–Wallis [15]
- Mann–Whitney U [16]

Seasonality Detection

- Monthly ranking metrics (mean, Sharpe [17], win ratio)
- Z-score entry/exit framework [18]
- Cumulative return by day-of-year
- Year-on-year consistency [19]
- Bootstrap CIs [20]
- Robust ANOVA (trimmed means) [21]
- Permutation test [22]

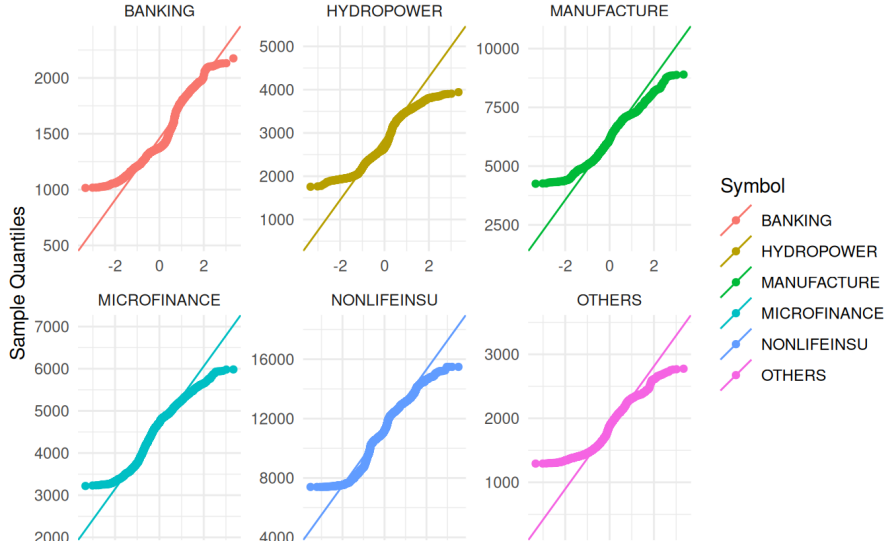
Regression: Simple linear trend and autoregressive model ($R_t = \beta_0 + \beta_1 R_{t-1} + \beta_2 \text{Sector} + \varepsilon_t$) [23].

EDA – Boxplot of Closing Prices



EDA – Q-Q Plots

Q-Q Plot of Close Price by Symbol



Distribution of Daily Returns

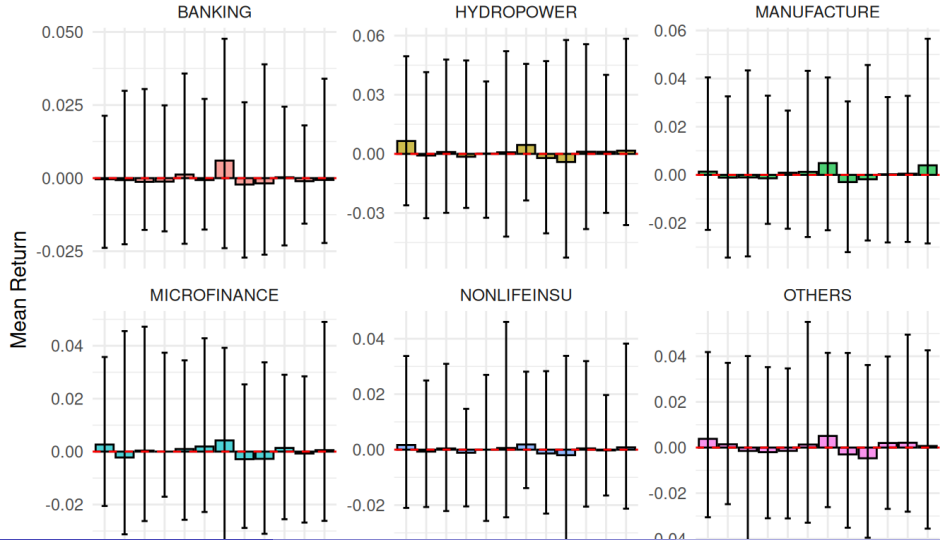
Table: Distributional characteristics of daily returns by sector

Sector	Mean	SD	Skewness	Excess Kurtosis
BANKING	-0.000117	0.0132	1.140	8.24
HYDROPOWER	0.000770	0.0209	0.615	4.44
MANUFACTURE	0.000458	0.0162	0.798	4.83
MICROFINANCE	0.000365	0.0164	0.802	6.58
NONLIFEINSU	0.000723	0.0114	0.976	9.88
OTHERS	0.000376	0.0184	0.638	4.87

- All sectors: positive skewness and high excess kurtosis $\Rightarrow$ fat tails [24].
- Shapiro–Wilk test rejects normality for all sectors ($p < 10^{-12}$) [11].

EDA – Monthly Average Returns with Bootstrap CIs

Monthly Average Daily Returns with 95% Bootstrap CIs



EDA – Return Density & Heatmap

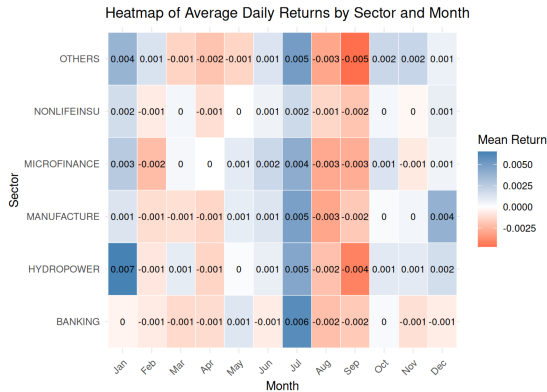
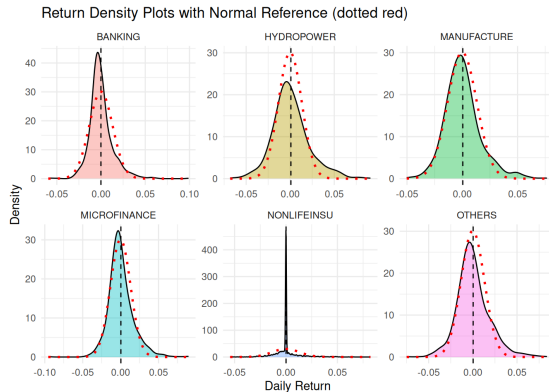


Figure: Density plots with normal reference (dotted red).

Figure: Heatmap: blue = positive, red = negative returns.

- Density plots confirm heavy tails and peakedness vs. normal.
- Heatmap highlights July as best, Aug–Sep as worst for all sectors.

EDA – Cumulative Returns & Price Distribution

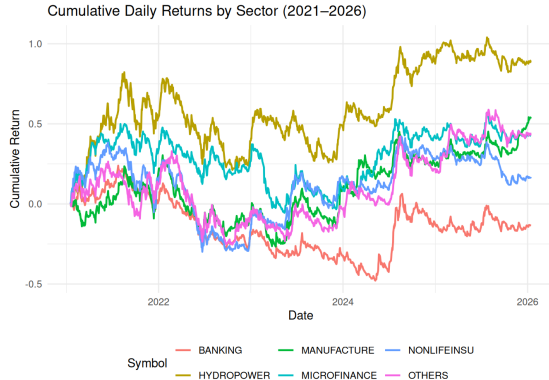


Figure: Cumulative daily returns (2021–2026).

- Hydropower shows strongest cumulative gain; Banking more modest.
- Histograms reveal right-skewed price distributions across sectors.

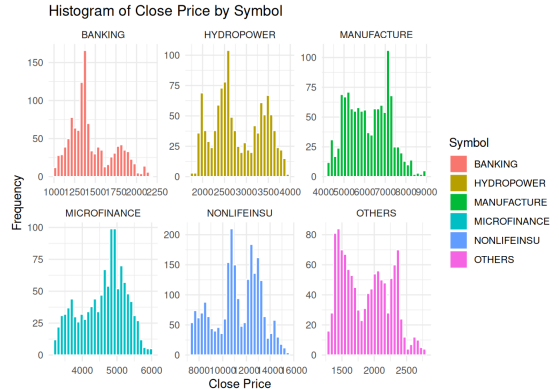


Figure: Histogram of closing prices by sector.

Month and Sector Effects – Statistical Tests

- **Chi-square test** [12]: Return direction (Gain/Loss) vs. month: $\chi^2(11) = 92.578, p = 5.2 \times 10^{-15} \Rightarrow$ strong dependence.
- **One-way ANOVA (month)** [13]: $F = 10.46, p < 2 \times 10^{-16} \Rightarrow$ significant monthly differences.
- **One-way ANOVA (sector)**: $F = 0.497, p = 0.779 \Rightarrow$ no significant sector differences.
- **Kruskal–Wallis (sector)** [15]: $H = 14.798, p = 0.0113 \Rightarrow$ weak sector effect.
- **Mann–Whitney (Banking vs Others)** [16]: $p = 0.8648 \Rightarrow$ no difference.

Takeaway

Strong evidence for **monthly seasonality**; little evidence of persistent sectoral differences.

Best and Worst Months

Two Best Months (by Mean Return)

Sector	Month	Mean	Win%
BANKING	Jul	0.00599	60.2
BANKING	May	0.00123	46.4
HYDROPOWER	Jan	0.00653	59.4
HYDROPOWER	Jul	0.00453	54.0
MANUFACTURE	Jul	0.00488	62.8
MANUFACTURE	Dec	0.00396	55.7
MICROFINANCE	Jul	0.00423	56.6
MICROFINANCE	Jan	0.00267	47.5
NONLIFEINSU	Jul	0.00187	26.5
NONLIFEINSU	Jan	0.00168	26.6
OTHERS	Jul	0.00509	60.2
OTHERS	Jan	0.00382	52.5

Two Worst Months

Sector	Month	Mean	Win%
BANKING	Aug	-0.00218	37.5
BANKING	Sep	-0.00179	36.7
HYDROPOWER	Sep	-0.00412	36.7
HYDROPOWER	Aug	-0.00210	40.4
MANUFACTURE	Aug	-0.00299	42.3
MANUFACTURE	Sep	-0.00184	35.6
MICROFINANCE	Aug	-0.00287	36.5
MICROFINANCE	Sep	-0.00274	38.9
NONLIFEINSU	Sep	-0.00199	18.9
NONLIFEINSU	Aug	-0.00136	18.8
OTHERS	Sep	-0.00469	32.2
OTHERS	Aug	-0.00300	31.7

July dominates as the best month; August–September worst across all sectors.

Z-Score Entry/Exit Framework

Table: ENTRY signals for Banking ($Z < -1.5$)

Month	Freq.	Avg Z
January	4	-2.24
February	4	-2.00
March	3	-1.90
April	4	-1.92
May	6	-2.03
June	5	-1.77
July	5	-1.84
August	5	-2.61
September	4	-2.10
October	4	-1.83

- Z-score: $Z_{s,m,t} = \frac{R_{s,m,t} - \mu_{s,m}}{\sigma_{s,m}}$
- $|Z| > 1.5$ flags extreme events [18].
- $Z < -1.5$: oversold $\Rightarrow$ ENTRY signal.
- $Z > +1.5$: overbought $\Rightarrow$ EXIT signal.
- Average entry $Z \approx -2.0$ (two SD below monthly mean).
- Conservative threshold of ± 1.8 recommended to reduce false signals given fat tails.

Cumulative Return and Optimal Exit Day

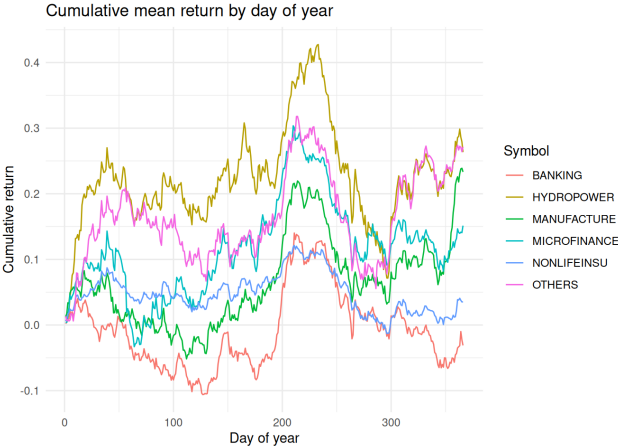


Table: Optimal exit day & cumulative return

Sector	Exit Day	Cum. Ret.
BANKING	210 (29 Jul)	0.113
HYDROPOWER	230 (18 Aug)	0.422
MANUFACTURE	365 (31 Dec)	0.202
MICROFINANCE	210 (29 Jul)	0.329
NONLIFEINSU	210 (29 Jul)	0.120
OTHERS	213 (1 Aug)	0.265

Figure: Cumulative mean return by day of year.

Late July–early August is the optimal exit for most sectors: Hvdronpower peaks later (day 230).

Year-on-Year Consistency

Table: Months with year-on-year consistency ≥ 0.80 [19]

Sector	Month	Positive/Total Years	Consistency
BANKING	July	5/5	1.00
HYDROPOWER	January	6/6	1.00
HYDROPOWER	July	4/5	0.80
HYDROPOWER	November	5/5	1.00
MANUFACTURE	January	5/6	0.833
MANUFACTURE	May	4/5	0.80
MANUFACTURE	July	4/5	0.80
MICROFINANCE	May	4/5	0.80
MICROFINANCE	July	4/5	0.80
NONLIFEINSU	May	4/5	0.80
NONLIFEINSU	July	4/5	0.80
NONLIFEINSU	November	4/5	0.80
OTHERS	January	5/6	0.833
OTHERS	July	4/5	0.80
OTHERS	November	5/5	1.00

Bootstrap Confidence Intervals

Table: 95% bootstrap percentile CIs for mean return in January [20]

Sector	Lower CI	Upper CI
NONLIFEINSU	0.00005	0.00334
MANUFACTURE	-0.00170	0.00456
BANKING	-0.00238	0.00178
OTHERS	0.00032	0.00730
MICROFINANCE	-0.00012	0.00584
HYDROPOWER	0.00263	0.01020

- Hydropower and Others: CIs entirely above zero $\Rightarrow$ significant positive mean.
- Banking's interval straddles zero — not statistically significant.
- Banking in August: CI $(-0.0048, 0.0004)$ includes zero, but direction uniformly negative.

Robust ANOVA and Permutation Test

Robust ANOVA (20% trimmed means) [21]

$F_t = 12.2519, p < 0.001$, effect size $\hat{\eta} = 0.16$ (95% CI [0.13, 0.19]) — moderate-to-large.

- Trimming removes extreme observations $\Rightarrow$ resistant to outliers and heterogeneity.
- Confirms month effect is **not** driven by a few extreme returns.

Permutation Test (5000 shuffles) [22]

$p < 2.2 \times 10^{-16}$ — strongest distribution-free evidence of seasonality.

- No distributional assumptions; provides the most robust validation.
- Together with robust ANOVA: monthly seasonality is a **genuine** feature, not an artifact.

Two-Way Interaction ANOVA & Summary of Key Findings

Two-Way ANOVA [14]

- **Sector:** $F = 0.502, p = 0.775$ — NS.
- **Month:** $F = 10.44, p < 2 \times 10^{-16}$ — **highly significant.**
- **Sector $\times$ Month:** $F = 0.702, p = 0.954$ — NS.

⇒ Seasonal pattern is **uniform** across sectors.

Key Findings

- **July** is the premier month: highest returns, win ratios $>60\%$, perfect consistency (Banking).
- **Aug–Sep** are avoidance windows: negative returns, low win ratios, negative Sharpe ratios.
- **Optimal exit:** day 210–213 (late Jul–early Aug); Hydropower later (day 230).
- **Z-score:** avg entry $Z \approx -2.0$; threshold ± 1.8 recommended.
- **Reliable:** July, January, November consistently positive across years.

Conclusion

- ❶ **Strong monthly seasonality** confirmed by multiple tests ($p < 0.001$).
- ❷ **July is the optimal entry month** — highest mean returns, win ratios, and consistency.
- ❸ **August–September are avoidance windows** — uniformly negative returns.
- ❹ **Optimal exit:** late July–early August; Hydropower later, Manufacturing requires caution.
- ❺ **Z-score framework** provides actionable signals; conservative threshold recommended.
- ❻ **Sector-invariant pattern** — a single seasonal calendar works for all sectors.

Practical Implication

These findings challenge weak-form efficiency [3, 2] and provide evidence-based timing rules for NEPSE investors [4, 5].

References I



B. Shrestha.

Seasonal anomalies in emerging markets: evidence from nepse.

Journal of Nepalese Business Studies, 15(1):45–60, 2023.



P. Joshi.

Broad market indices of nepal stock exchange: Testing of efficient market hypothesis.

Academia Journal of Research & Innovation, 1(1):60–70, 2025.



A. Basnet.

Testing weak-form market efficiency in nepse: a comprehensive review.

Nepalese Journal of Finance, 4(1):22–38, 2022.



R. Gurung and P. Subedi.

Fiscal year and festive effects on market price movements: A guide to trading strategies at nepal stock exchange.

Pravaha, 30(1):20–30, 2024.

References II



S. Adhikari and R. Koirala.

Calendar effects and trading strategies in south asian stock exchanges.

Asian Journal of Finance & Economics, 12(2):89–104, 2024.



S. Sharma and B. Thapa.

Investor sentiment and calendar anomalies: evidence from nepal.

South Asian Journal of Business Studies, 10(3):301–318, 2021.



John W. Tukey.

Exploratory Data Analysis.

Addison-Wesley, 1977.



John M. Chambers, William S. Cleveland, Beat Kleiner, and Paul A. Tukey.

Graphical Methods for Data Analysis.

Wadsworth, 1983.



D. N. Joanes and C. A. Gill.

Comparing measures of sample skewness and kurtosis.

The Statistician, 47(1):183–189, 1998.

References III

 Lawrence T. DeCarlo.
On the meaning and use of kurtosis.
Psychological Methods, 2(3):292–307, 1997.

 S. S. Shapiro and M. B. Wilk.
An analysis of variance test for normality (complete samples).
Biometrika, 52(3/4):591–611, 1965.

 Karl Pearson.
On the criterion that a given system of deviations from the probable in the case of a correlated system of variables is such that it can be reasonably supposed to have arisen from random sampling.
Philosophical Magazine, 50(302):157–175, 1900.

 Ronald A. Fisher.
Statistical Methods for Research Workers.
Oliver and Boyd, 1925.

References IV



Michael H. Kutner, Christopher J. Nachtsheim, John Neter, and William Li.
Applied Linear Statistical Models.
McGraw-Hill, 5th edition, 2005.



William H. Kruskal and W. Allen Wallis.
Use of ranks in one-criterion variance analysis.
Journal of the American Statistical Association, 47(260):583–621, 1952.



Henry B. Mann and Donald R. Whitney.
On a test of whether one of two random variables is stochastically larger than the other.
Annals of Mathematical Statistics, 18(1):50–60, 1947.



William F. Sharpe.
Mutual fund performance.
Journal of Business, 39(1):119–138, 1966.



Ruey S. Tsay.
Analysis of Financial Time Series.
Wiley, 3rd edition, 2010.

References V



Josef Lakonishok and Seymour Smidt.

Are seasonal anomalies real? a ninety-year perspective.

Review of Financial Studies, 1(4):403–425, 1988.



Bradley Efron and Robert J. Tibshirani.

An Introduction to the Bootstrap.

Chapman & Hall/CRC, 1994.



Rand R. Wilcox.

Introduction to Robust Estimation and Hypothesis Testing.

Academic Press, 4th edition, 2017.



Phillip Good.

Permutation, Parametric, and Bootstrap Tests of Hypotheses.

Springer, 3rd edition, 2005.



Norman R. Draper and Harry Smith.

Applied Regression Analysis.

Wiley, 3rd edition, 1998.

References VI



Rama Cont.

Empirical properties of asset returns: stylized facts and statistical issues.
Quantitative Finance, 1(2):223–236, 2001.

Thank You

Questions & Discussion